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Observations of Recurrent Laryngeal Nerve Injury and Recovery Using a Rat Model

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Objectives/Hypothesis: To evaluate standardized recurrent laryngeal nerve (RLN) injuries using a rat model via minimally invasive transoral electromyography (ToL EMG) and histologic studies.

Methods: Forty-two female Sprague Dawley rats weighing 200 g to 250 g underwent crush injury to the right RLN using a calibrated pressure clip (0.61 N or 1.19 N) for 60 seconds. Following injury, serial ToL EMGs were performed on abductor and adductor laryngeal muscles during respiratory cycles and spontaneous vocal fold abduction on day 4 and then weekly for 6 weeks. Vocal fold motion associated with spontaneous respiration was graded from 0 to 4. Rats were sacrificed at different time points for histologic evaluation of injured nerves.

Results: EMG signals showed fibrillation potentials on day 4 in all experimental conditions. Crushed RLN, regardless of force, exhibited polyphasic potentials at 2 weeks postinjury. Normal motor unit potentials and recruitment patterns were observed in EMG signals at 4 weeks for all 0.61 N clip animals. Six weeks following crush injury, motor unit potentials having normal appearance were observed in most animals. Synkinetic EMG signals were observed at 5 weeks and 6 weeks in the 1.19 N clip animals. Endoscopic evaluation of vocal fold mobility was consistently normal at 6 weeks only following 0.61 N crush injury.

Conclusions: This model is useful to simulate intraoperative RLN injuries and to better understand the electrophysiologic events during nerve recovery. The severity of injury to the RLN dictates histologic, neurologic and functional recovery of the laryngeal motor system. This model is useful to evaluate the efficacy of systemic and local neurotropic agents in the treatment of RLN injury.

Key Words: RLN injury, rat model, histology, electromyography.

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INTRODUCTION

The recurrent laryngeal nerve (RLN) is sometimes injured during surgeries of the head and neck region. Prognosis of functional recovery is challenging and often innacurate.[1] Recovery from RLN injury is dependent on the mode and extent of injury. As our knowledge of the biology of peripheral nerve regeneration increases, there has been increasing interest in developing models of nerve regeneration that are pertinent to specific clinical questions. Succesful return of normal function is variable. Neurotrophic factors are promising but, so far, have inconsistencies in the results. A highly focused study with predictable results is needed to effectively evaluate trophic factors. Various experimental animal models have been used to study peripheral nerve injury and functional recovery.[2–6] Recently, several studies have used the rat larynx to evaluate potential agents that promote recovery of the RLN following injury.[7–11]

Aneurysm clips provide a consistent and reliable crush injury to peripheral nerves with a calibrated force that is applied to an area over a specific time.[12,13] A standard method of compression will serve to ameliorate discrepancies among research teams when studying crush injury of the RLN. The aim of the present study was to evaluate the regeneration and functional recovery of RLN following controlled crush injury. This model is intended to simulate one type of intraoperative RLN injury and to facilitate understanding of the histological and myoelectrical events that are observed during nerve recovery.

MATERIALS AND METHODS

Animals

Forty-two Sprague Dawley rats weighing 200 g to 250 g were used in the present study. The animals were divided into two groups. The first group underwent crush injury using a

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commercially available Sugita calibrated 0.61 N aneurysm clip (Mizuho Ikakogyo, Tokyo, Japan), the second group underwent crush injury using a 1.19 N aneurysm clip (Mizuho Ikakogyo). Both aneurysm clips were 1.4 mm in width. Animal procedures were approved by the institutional review board of New York Medical College and performed in facilities approved by the National Academy of Sciences and the National Society for Medical Research. Humane care was provided for these animals and all institutional and national guidelines were observed.

Surgery

Animals were sedated with isoflurane and then anesthetized with an intramuscular injection of a mixture of Ketaset and Rompun (1:1). A final dosage of 45 mg ketamine/100 g body weight and 0.9 mg/100 g body weight of xylazine plus 1 mg/kg acepromazine maleate produced sufficient anesthesia.

With animals placed in a supine position in a stereotactic apparatus, a vertical midline neck incision was performed, and the strap muscles were separated. The right recurrent laryngeal nerve was crushed in all animal groups and the left side was used as control. The RLN was carefully identified in the tracheoesophageal groove under a dissecting microscope. In the first group of animals, crush injury was performed at the level of the seventh tracheal ring with a 0.61 N closing force for exactly 60 seconds. The second group underwent an identical procedure using a Sugita 1.19 N closing force. Strap muscles and overlying fascia were closed with 4-0 chromic suture and the skin reapproximated with 5-0 nylon sutures.

Transoral Evaluation of Vocal Fold Movement EMG

Functional behavior was studied using transoral laryngeal electromyography (ToL EMG) obtained during spontaneous respiratory cycles and vocal fold adduction under anesthesia. The methodology of ToL EMG has been published in our earlier paper,[14] with salient details provided here. The animals were placed supine on a modified stereotactic operating table with a 15° incline in a Faraday cage recording room. The tongue was retracted using a 3-0 silk suture placed midline in the anterior one third of the tongue and suspended. A 0° Storz operating endoscope equipped with an epiglottis elevator was transorally inserted to visualize the endolarynx and provide excellent visualization of the glottal region. Vocal fold movements observed via transoral endoscopy were independently graded by two investigators using a continuous scale from 0 to 4 prior to performing each ToL EMG. Scoring of vocal fold movement and position was guided by the following criteria: 0 = no vocal fold movement; 1 = slight vocal fold movement; 2 = <50% abduction of vocal folds; 3 = >50% abduction, unsmooth vocal fold movement; 4 = vocal fold movement and position that is indistinguishable from normal. Scores from the two investigators were combined into a single score by assigning the average value.

Transoral laryngeal EMG was performed on postoperative day 0, 4, 7, 14, 28, 33, and 42, recording successively from adductor muscles and posterior cricoarytenoid muscles on both the injured and control side. Muscle identification followed the scheme published by Inagi et al.[15] Myoelectric recordings from posterior cricoarytenoid muscles were obtained during phasic respiratory cycles. Recordings from adductor muscles were obtained during periods of spontaneous adduction. All myoelectric activity was spontaneous and unevoked. Myoelectric signals were detected using a 25-gauge, 45-mm quadrifilar needle electrode (Delsys, Boston, MA). The quadrifilar electrode has four 50-μm diameter selective surfaces located along cannula side ports that are spaced 200 μm apart in a square pattern.[16] Bipolar EMG signals were recorded from two of the four electrode detectors. Assuming that diameters of rat laryngeal muscle fibers are approximately 25 μm,[7] detectors of the quadrifilar electrode, therefore, spanned multiple muscle fibers while providing an increased selectivity of individual motor unit action potential morphologies over concentric needle electrodes. We also tested clinically-available bipolar concentrilc needle and monopolar needle electrodes for obtaining EMG recordings from the rat larynx; however, the quadrifilar electrode provided the most consistent and reliable recordings for the rat model. An ear clip electrode served as reference for the EMG recordings.

Myoelectric signals were routed to a preamplifier module located at the needle head and then to high-gain differential amplifiers located nearby (in-house construction). Myoelectric signals were bandpass filtered from 15 Hz to 1,000 Hz (Krohn-Hite Model 3360, Brockton, MA) and then digitized at a depth of 12 bits and a rate of 20 kilosamples per second (Data Translation Model DT2821G, Marlboro, MA). Digitized data were stored to the internal disc drive of a portable computer in real time (Dolch Computer Systems Model LPAC-PT, Freemont, CA). EMG signals were displayed, analyzed, and compared using Matab software (The Mathworks, Release 2007b, Natick, MA).

Myoelectric activity was also monitored visually using a digital real-time oscilloscope (Tektronix TDS210, Tektronix, Beaverton, OR) and audibly by means of an amplifier/speaker unit connected in parallel to the EMG data recoding system. Kinematic activity (e.g., vocal fold movement and position) was observed visually by one investigator at a given time by means of ToL endoscopy. In some experiments, kinematic activity was observed by all investigators by means of a video monitor that received its signal from a solid-state color camera (Richard Wolf S5366.40, Richard Wolf GmbH, Knittlingen, Germany) mounted on the eyepiece of the endoscope. Simultaneous recordings of kinematic and myoelectric activities were achieved for some experiments by feeding the audio/video signals to a VHS tape recorder (Panasonic AG-6300MD, Panasonic, Knoxville, TN). The audio/video signals were digitized to AVI files by a computer workstation following the experimental session. AVI files facilitated postexperimental review and discussion. This technique helped to confirm judgments that were made during the experimental session, including incidences of synkinesis.

Histology

Following ToL EMG, animals were sacrificed with an intraperitoneal injection of lethal dose of phenobarbital at postoperative day 4, 7, 14, 21, 28, 33, and 42. Immediately thereafter, recurrent laryngeal nerves were harvested. The recurrent laryngeal nerve was identified in the tracheoesophageal groove, and the crushed segment was located at the seventh tracheal ring. The nerve was harvested and specimens were taken from two segments above and below the injury. The contralateral nerves served as controls and were harvested at the level of the seventh tracheal ring. All specimens were immediately fixed in a mixture of 1% glutaraldehyde and 4% paraformaldehyde in 0.1 M pH 7.2 phosphate buffer and were postfixed for 1 hour in 1% osmium tetroxide in 0.1 M phosphate buffer and dehydrated in graded ethanol and embedded in epoxy resin in standard fashion. Thin sections of 1 μm were cut with glass knives on a Sorvall ultramicrotome and stained with aqueous 1% toluidine blue.

For electron microscopy, 80-nm thick sections were cut with a diamond knife on a Sorvall ultramicrotome. These sections were mounted on 2 $\times$ 1 mm colloidin-coated grid that allowed visualization of the entire cross-section of the nerve.

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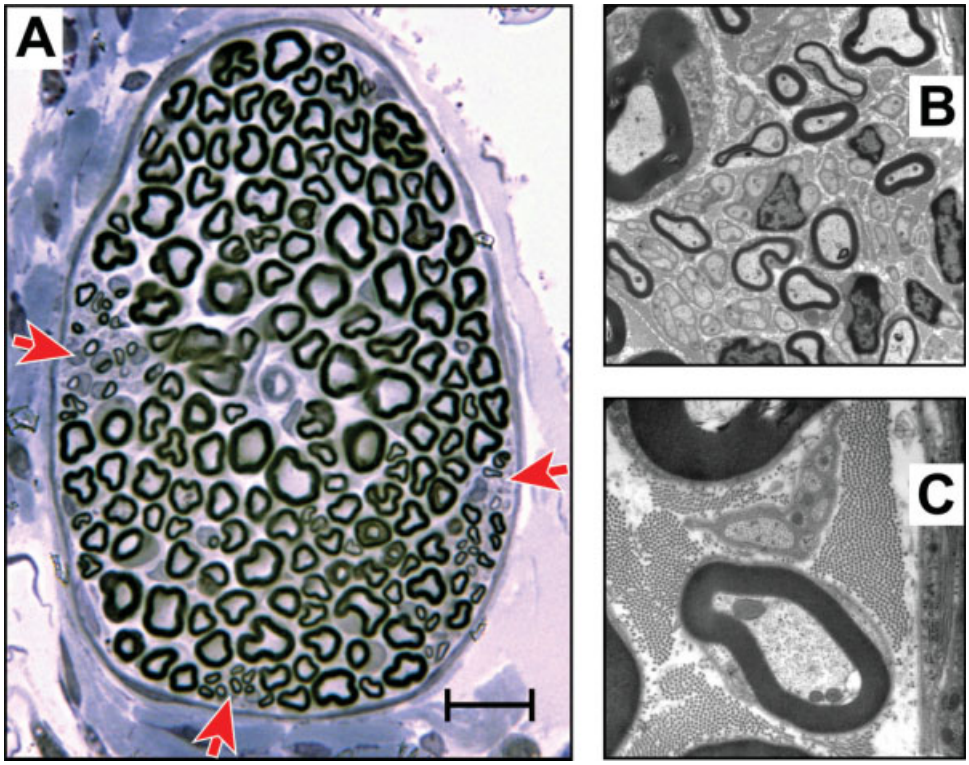


Fig. 1. (A) One-micron plastic section visualized under light microscope of a toludine blue-stained recurrent laryngeal nerve cross-section of normal rat. Red arrows point to less myelinated and some unmyelinated axons as well. Scale bar is 50 μm. Electron micrographs of normal recurrent laryngeal nerve cross-section showing myelinated and unmyelinated axons and Schwann cells at original magnifications of (B) $\times$15,000, and (C) $\times$45,000.

Sections were visualized on Zeiss EM 900 (Carl Zeiss AG, Oberkochen, Germany) transmission electron microscope at varied magnifications ranging from 150 to 50,000.

RESULTS

Histologic Results

While using the 0.61 N clip, the histologic results were variable. There were compressed axons in some cases and degenerating axons in others. In spite of variable histological outcome, electrophysiological data were consistent. Hence it was deemed necessary to use higher force to induce more uniform histologic changes in the RLN. Because the histologic results of the study using a 0.61 N clip force were inconsistent, they will not be described further for the sake of focus. Results of RLN histology and electrophysiology appearing below are associated with the 1.19 N clip placed for 60 seconds.

One micron plastic section of a control right RLN viewed under lower power revealed that the majority of axons were heavily myelinated and of large caliber. See Fig. 1A. Scattered among these axons were smaller caliber axons that were also heavily myelinated. There were two or three groups of very small caliber axons that were mostly confined to the peripheral portion of the nerve close to the perineurium. These groups are indicated by the red arrows in Fig. 1A. In total there were 300 $\pm$ 29 (n = 8) axons in RLN: 287 $\pm$ 10 were myelinated axons whereas 13 $\pm$ 5 were unmyelinated axons. Unmyelinated axons may represent newly added axons. The segregated axonal grouping may also be indicative of a certain order in the nerve and may represent the afferents to the brain centers. Because most unmyelinated and less myelinated axons in other mammals represent pain and temperature fibers, it is conceivable that these axons are destined to the nucleus tractus solitarius.

An electron micrograph of normal RLN cross-section is shown in Figure 1B. The image shows myelinated and unmyelinated axons. Astrocytes and Schwann cells are also clearly visible. A higher magnification of the central area of the nerve appears in Fig. 1C, showing myelinated axons surrounded by Schwann cells and collagen fibrils.

Four days after the right RLN compression using the 1.19 N clip, nerve harvested 2 mm distal to the site of injury showed remarkable changes in the axonal profiles. In addition to the breakup of myelin, these were numerous vacuoles, dilated axons, and presence of large numbers of glia and macrophages (Fig. 2A). Intact myelinated axons were rare. At day 7 postcompression, distal to the injury there were degenerating profiles of all axons in all animals (n = 4) (Fig. 2B). Large number of glia and macrophages were evident. No regenerating axonal growth was noted at this stage. In the proximal portion of the RLN, harvested 2 mm proximal to the site of injury, axonal profiles were almost normal except for a few degenerating axons that may represent sensory afferent axons.

At day 14 postcompression, the distal portion of the nerve showed thinly myelinated and more unmyelinated axons than controls. Macrophages and other glia were still noticeable except that their number was appreciably reduced when compared to that seen at day 7 after the nerve crush (Fig. 2C). At day 21 postcompression, the

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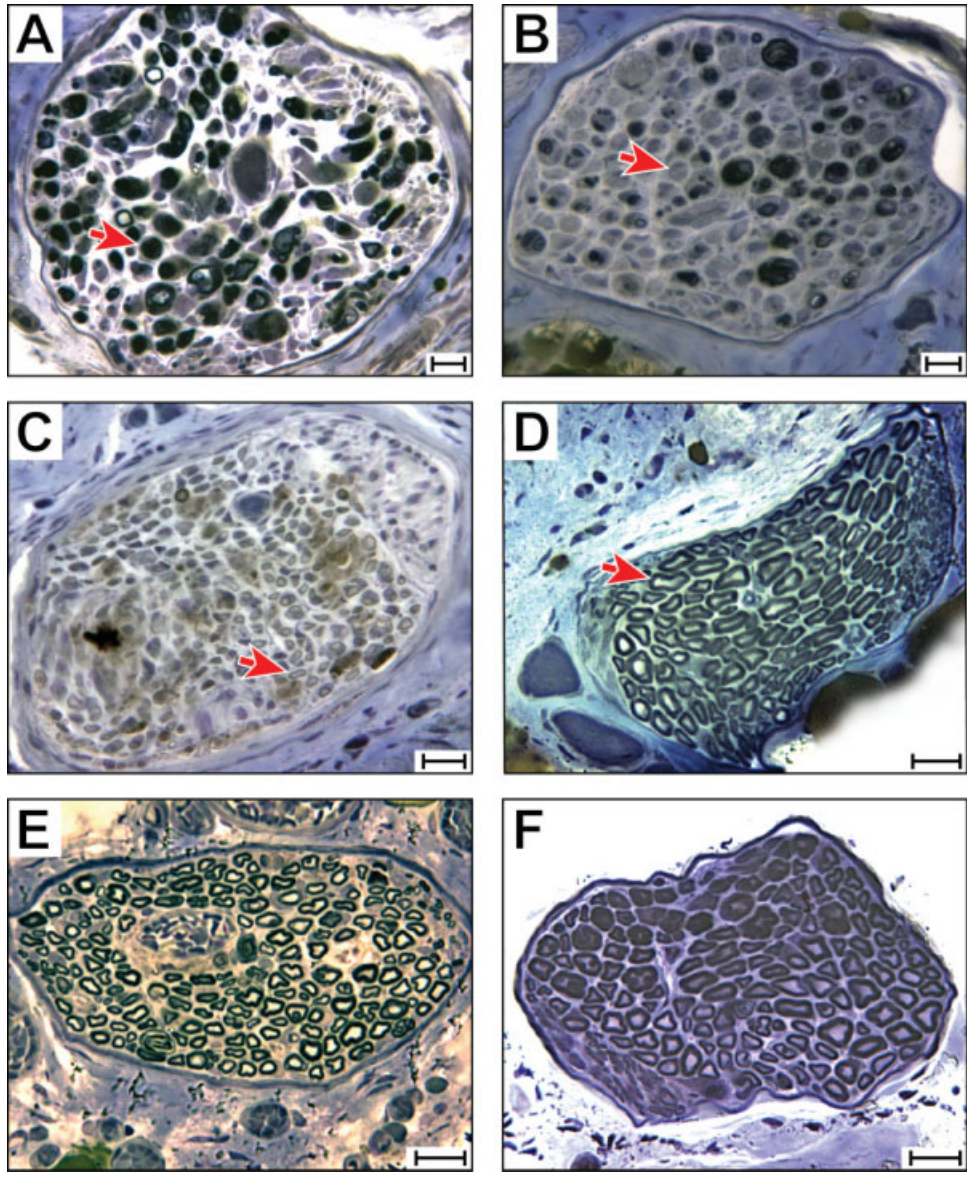


Fig. 2. Representative sections of rat recurrent laryngeal nerves distal to injury at different time points: (A) 4 days, (B) 7 days, (C) 14 days, (D) 21 days, (E) 33 days, and (F) 42 days after the injury. Scale bars are each 50 μm. Arrows point to the breakdown in myelin profiles at day 4 in (A), degenerating axonal profiles at day 7 in (B), thinly myelinated axons at day 14 in (C), and thickly myelinated axons in (D). (E) and (F) show nearly normal axonal profiles.

axonal profiles showed large diameter axons, which were heavily myelinated. The number of axons with relatively less myelin or still unmyelinated remained large. These axons were mostly confined to the periphery of the nerve near the perineurium. Occasional macrophages were still encountered (Fig. 2D).

At day 33 postcompression, the number of less myelinated axons was small. In all aspects, the nerve appeared similar to the normal nerve except some axons were less myelinated than normal (Fig. 2E). At day 42 postcompression, axonal profiles in the cross section were similar to that seen in the adult normal nerve. Nearly all large diameter axons were heavily myelinated, though to different degrees (Fig. 2F). The pattern of histologic nerve regeneration described above correlates well with the electrophysiological data described below.

Electron micrographs showing RLN cross-section at 7 days and 14 days following the nerve crush are shown if Figure 3. At 7 days (Fig. 3A), most axons show degenerating myelin debris and vacuoles in the axons. At 14 days after nerve crush (Fig. 3B), some myelin debris is still present; however, myelinating axons are evident and surrounded by Schwann cells. Original magnification for micrographs in Figs. 3A and 3B is $\times$9,000.

Electromyographic Results

The ToL EMG recording appearing in Figure 4A was obtained from left posterior cricoarytenoid (PCA) muscle during a series of four respiratory cycles. The EMG signal in Figure 4B was obtained when recording from the left bundle of adductor muscles[2] during spontaneous adductive activity (e.g., medial approximation, cough, and swallow). The EMG electrode typically detected two or three motor units for the noncompromised PCA muscle during respiration, and three to four motor units for adductor muscles during spontaneous

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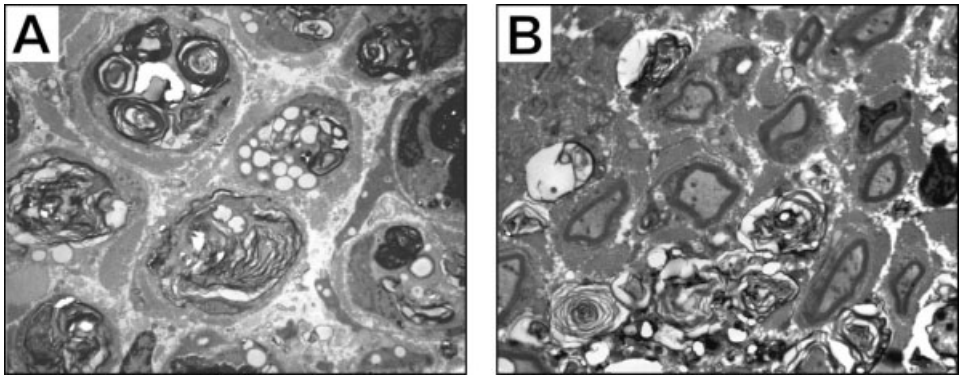


Fig. 3. Electron micrographs showing recurrent laryngeal nerve cross-section at (A) 7 days and (B) 14 days following the nerve crush. At 7 days (A), most axons show degenerating myelin debris and vacuoles in the axons. At 14 days after nerve crush (B), some myelin debris is still present; however, myelinating axons are evident and are surrounded by Schwann cells. Original magnification is ×9,000 for each micrograph.

laryngeal activity. Identification of motor unit number was obtained from our EMG decomposition procedure published previously.[14] The recordings appearing in Figure 4 were obtained prior to right-side RLN injury. Features of normal myoelectric physiology (as characterized by morphology, motor unit number, and recruitment) in control side muscles appeared unaffected by RLN injury to the contralateral side. No animals exhibited respiratory phasic myoelectric activity in adductor muscles on the uninjured side.

Breathing patterns were not notably altered by injuries; however, hyperadduction was observed in some animals postinjury (swallow and cough), which was likely associated with a laryngeal response to injury as observed in the supine position (aspiration). No animals were lost between experimental procedures, and specimens appeared vigorous and otherwise healthy throughout the study. Confirmation of RLN injury was randomly verified in 20 animals on the same day of injury via vocal fold immobility observed by transoral endoscopy (movement score = 0) and by myoelectric silence of EMG recordings in PCA and adductor muscles.[14] EMG and visual evaluation of RLN recovery after the 0.61 N crush injury revealed findings consistent with the milder RLN injury noted in our earlier report.[14] As discussed in previous paragraphs, histologic findings associated with the 0.61 N clip were heterogeneous due to inconsistent injury, and, therefore, electrophysiologic findings of this article will focus upon 1.19 N crush results.

Figure 5A shows a ToL EMG recording obtained on day 4 of the study for a 1.19 N crush specimen. The expanded view shows a repetitive action potential (AP) dominating the recording field accompanied by far-field potentials having duration, shape, and regularity that are typical of single-fiber fibrillations associated with nerve injury. All myoelectric activity appeared absent of any phasic pattern that could be associated with respiration. Fibrillation potentials were observed in all experimental conditions on day 4 for 92% of animals, with the remaining specimens exhibiting APs having complex morphology that were indicative of a recovery stage, as described below. Figure 5B shows a PCA EMG recording obtained on day 7. Myoelectric activity is characterized by an increase in spontaneous fibrillation potentials and positive sharp waves. Myoelectric activity appears to be disassociated from respiration, as noted in earlier stages. Movement scores averaged 0.25 for day 4 recordings and 1.25 for day 7 recordings.

Figure 5C is representative of PCA EMG recordings obtained at day 14. Although some degree of spontaneous firings persist, respiratory phasic activity has largely returned and complex motor unit action potentials (MUAP) have emerged that are consistent with axonal reconnections to multiple muscle fibers. Figure 6A illustrates PCA EMG recordings at day 21, showing a refinement and narrowing of MUAP shape and continued subjugation of disassociated APs. In this instance, the recording field is dominated by a single motor unit having features that are consistent with intermediate recovery. During abduction, vocal fold movement on the injured side was typically more abrupt in its lateralization than the contralateral vocal fold, which evidenced a longer deceleration phase and smoother overall movement. Movement scores averaged 1.75 for day 14 recordings and 2.25 for day 21 recordings.

A PCA EMG recording from an animal at day 33 is illustrated in Fig. 6B. Multiple motor units have emerged having complex regeneration shapes with increased duration. Smoother vocal fold movement

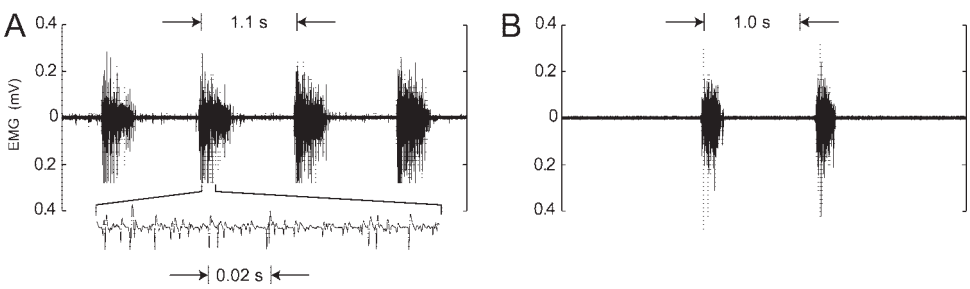


Fig. 4. Electromyograph recordings from two laryngeal muscles of a normal rat. (A) The posterior cricoarytenoid muscle during respiration. (B) A group of adductor muscles during spontaneous activation.

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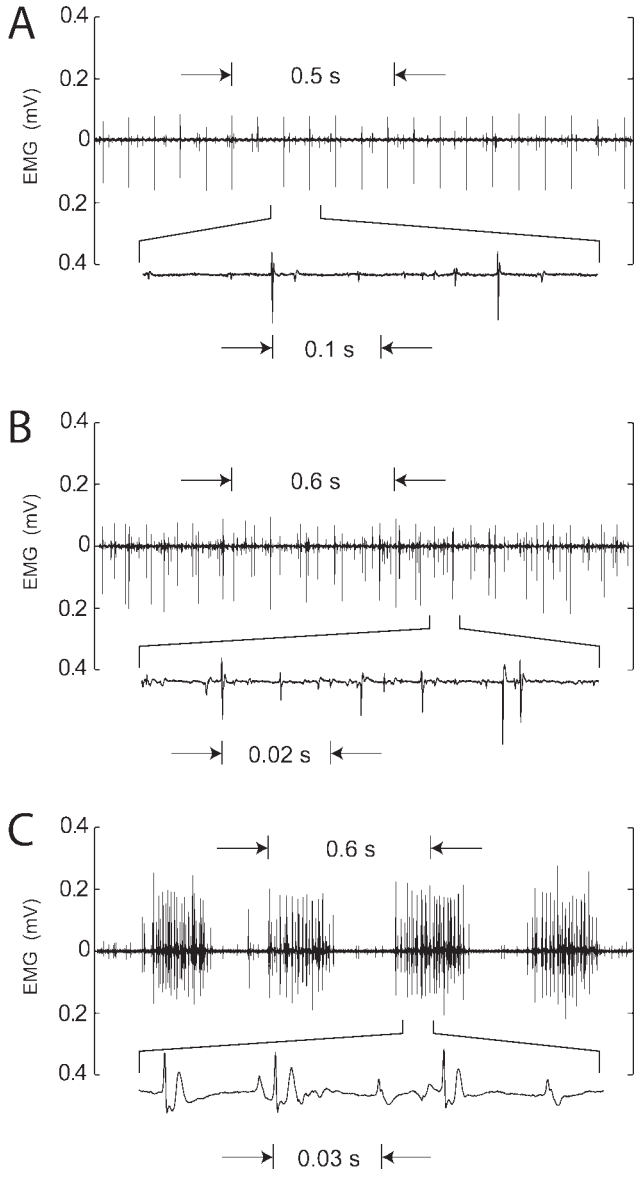


Fig. 5. Electromyograph recordings from the posterior cricoarytenoid muscle of rat following injury at different time points. (A) 4 days, showing muscle potentials having characteristics consistent with fibrillation. (B) 7 days, showing increased fibrillation and positive sharp wave potentials. (C) 14 days, evidencing potentials that are consistent with the beginning stages of nerve recovery and return of phasic motor drive.

observed at day 33, as compared to earlier stages, appeared to be associated with refinements of motor unit recruitment and decruitment that was evidenced by the EMG data. Figure 6C is representative of PCA EMG recordings obtained at day 42 for the group of four specimens, where MUAPs have assumed shapes characteristic of normal morphology. Movement scores averaged 3.0 for day 33 recordings and 3.5 for recordings made on day 42.

Myoelectric activity present in the abductor muscles on adduction of the contralateral muscle or myoelectric activity in the adductor muscles on abduction of the contralateral muscle was considered synkinetic activity for the purpose of this study. A judgment of synkinesis required consensus by a minimum of two investigators, but was not graded to a specific metric, such as severity. Beginning with day 33 EMG recordings, we noted synkinesis in adductor muscles for two animals when observing respiratory phasic myoelectric activity that was coincident with vocal fold abduction observed by endoscopy. In one animal, sacrificed on day 33, vocal fold movement was scored as 2.0, whereas in the other animal sacrificed on day 42 vocal fold movement was scored as 3.0. Similarly, synkinesis of abductor muscle was noted in one animal from the 33 day group. Synkinesis in posterior cricoarytenoid muscle required multiple incidences of adduction (swallowing and/or coughing) rather than a single observation.

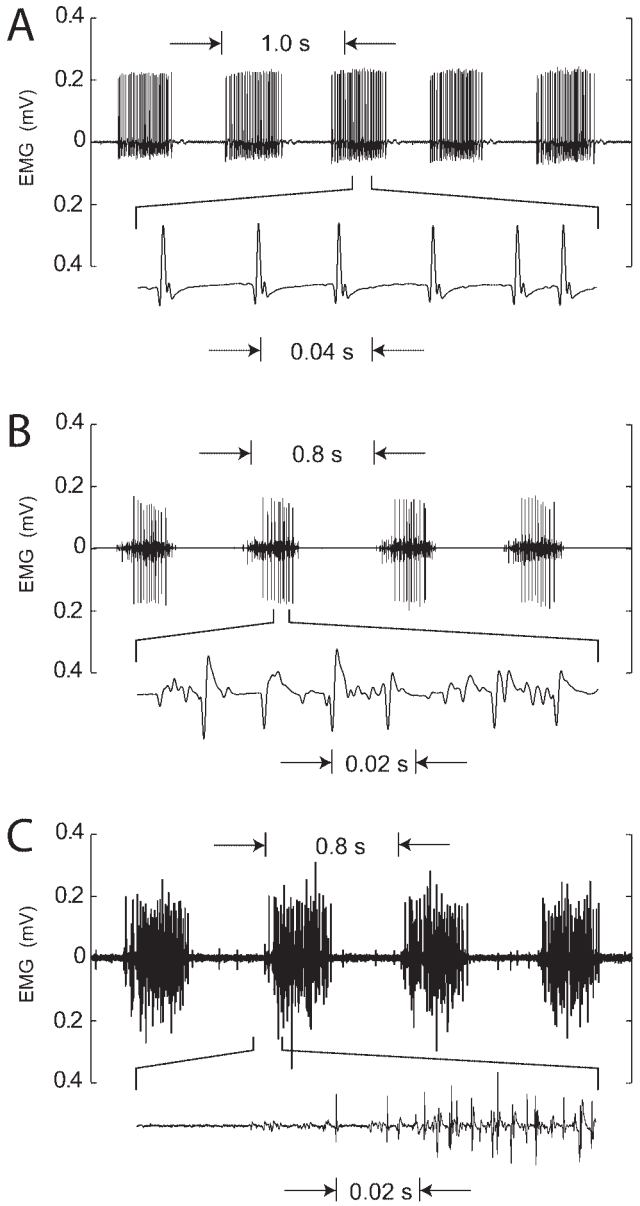


Fig. 6. Electromyograph recordings from the posterior cricoarytenoid muscle of rat following injury at (A) 21 days, (B) 33 days, and (C) 42 days, demonstrating an increasing return of neuromotor activity and richness of motor unit activations.

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DISCUSSION

We have developed an animal model that closely evaluates the injury and regeneration of rat recurrent laryngeal nerve. Transoral electromyography provides investigators the ability to evaluate electrophysiologic changes that correspond to standardized injuries in an intact animal over a prolonged period. We recently published our ToL EMG method of using an operating endoscope to visualize the rat endolarynx and perform serial electromyography after a controlled crush injury to the recurrent laryngeal nerve.[14] Our minimally invasive ToL EMG protocol offers several advantages over traditional methods while providing electrophysiologic data that can be used to explore diagnostic and prognostic signatures of specific injuries and the stage of recovery. Moreover, vocal fold mobility for evaluation of functional outcome and incidence of synkinesis can be evaluated. Electrophysiological examination of laryngeal muscles using stimulated compound muscle action potentials does not have clinical applicability, whereas ToL EMG data is collected during spontaneous, unevoked respiratory cycles that can provide clinically relevant data. In the current study, we correlated the electrophysiologic changes with histologic evidence of nerve injury and regeneration at different time points. This was performed after varying degrees of standardized injury to the RLN.

Several studies have used the rat larynx as a model to study nerve injury. However, none of these studies provided a standard method of crush injury. Such a standard should include the intensity and duration of crush injury. This is particularly important when attempting to elucidate the effects of exogenous agents on nerve recovery. Our histologic and electrophysiologic results indicate that a force of 1.19 N applied for 60 seconds to the rat recurrent laryngeal nerve results in complete Wallerian degeneration. Mori et al. recently reported on the efficacy of a novel drug T-588 after crush injury to the rat RLN.[10] They reported a statistically significant difference in recovery of vocal fold movement at week 4. In a similar study, however, Hydman et al.[11] reported no difference in nerve latency time that reflected degrees of myelination of the axons at 4 weeks after crush injury for both nimodipine-treated and untreated groups. Our data shows that functional recovery measured by vocal fold movement does not completely normalize after crush injury with a force of 1.19 N. Synkinesis was observed in three animals starting at 4 weeks after injury.

Functional recovery after RLN injury is dependent on the severity of injury. Crush injury to the RLN results in neuropraxia with a force of 0.61 N, whereas axonotmesis results when a crush injury is performed with a force of 1.19 N. Standardization of the mode of injury is necessary to understand the natural history of RLN recovery, whereas differences in functional outcome within a study population can be studied using the animal model and methods presented here.

Electromyography has been traditionally performed on experimental animals via invasive procedures to gain access to the laryngeal muscles. Our minimally invasive ToL EMG procedure provides several advantages over other recording methods while providing selective electrophysiologic data that can be used to explore diagnostic and prognostic signatures of specific injuries and the stage of recovery. Moreover, vocal fold mobility for evaluation of functional outcome and the course of synkinesis can be simultaneously evaluated. Electrophysiological examination using stimulated compound muscle action potentials does not have clinical applicability. However, ToL EMG data are collected during spontaneous, unevoked respiratory cycles, which can provide clinically relevant data.

CONCLUSION

The model and methods introduced in this article are intended to simulate intraoperative RLN crush injury and to better understand the electrophysiologic and histologic events occurring during nerve recovery. Standardized RLN injury using aneurysm clips in a rat model resulted in a consistent nerve injury and recovery that was followed longitudinally with transoral EMG, histologic evaluation, and movement evaluation. Electrophysiologic findings were found to correlate with histologic results of nerve degeneration and regeneration after injury

The severity of injury to the RLN dictates functional recovery of vocal fold motion. All animals with a 0.61 N force RLN crush recovered full vocal fold action at 7 weeks, whereas animals with the 1.19 N crush animals evidenced an average 3.5 out of 4.0 recovery score. No animals from the 0.61 N population evidenced synkinesis, compared to three animals from the 1.19 N crush group.

In future studies, we plan to expand the model to include additional RLN injuries, including increased severity of crush and transection. With enhanced elaboration of the animal model and better understanding of the regenerative processes, the studies will provide a platform to effectively pursue neuroregenerative agents following RLN injury that can be extended to clinical settings.

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BIBLIOGRAPHY

1. Sulica L, Blitzer A. Electromyography and the immobile vocal fold. *Otolaryngol Clin North Am* 2004;7:59–74.
2. Inagi K, Schultz E, Ford CN. An anatomical study of the rat larynx: establishing the rat model for neuromuscular research. *Otolaryngol Head Neck Surg* 1998;118:74–81.
3. Hinrichen CFL, Ryan AT. Localization of laryngeal motoneurons in the rat: morphologic evidence for dual innervations? *Exp Neurol* 1981;74:341–355.
4. Davis PJ, Nail BS. On the location and size of laryngeal motoneurons in cat and rabbit. *J Comp Neurol* 1984;230: 13–32.
5. Mu L, Yang S. An experimental study on the laryngeal electromyography and visual observation in varying types of surgical injuries to the unilateral recurrent laryngeal nerve in the neck. *Laryngoscope* 1991;101:699–708.

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6. Berkowitz RG, Sun QJ, Chalmers J, Pilowsky PM. Respiratory activity of the rat posterior cricoarytenoid muscle. *Ann Otol Rhinol Laryngol* 1997;106:897–901.
7. Flint PW, Downs DH, Coltrera MD. Laryngeal synkinesis following reinnervation in the rat. Neuroanatomic and physiologic study using retrograde fluorescent tracers and electromyography. *Ann Otol Rhinol Laryngol* 1991; 100:797–806.
8. Shiotani A, Nakagawa H, Flint PW. Modulation of myosin heavy chains in rat laryngeal muscle. *Laryngoscope* 2001; 111:472–477.
9. Shiotani A, O'Malley BW, Coleman ME, Flint PW. Human insulinlike growth factor 1 gene transfer into paralyzed rat larynx: single vs multiple injection. *Arch Otolaryngol Head Neck Surg* 1999;125:555–560.
10. Mori Y, Shiotani A, Saito K, et al. A novel drug therapy for recurrent laryngeal nerve injury using T-588. *Laryngoscope* 2007;117:1313–1318.
11. Hydman J, Remahl S, Bjorck G, Svensson M, Mattsson P. Nimodipine improves reinnervation and neuromuscular function after injury to the recurrent laryngeal nerve in the rat. *Ann Otol Rhinol Laryngol* 2007;116:623–630.
12. Sarikcioglu L, Ozkan O. Yasargil-Phynox aneurysm clip: a simple and reliable device for making a peripheral nerve injury. *Int J Neurosci* 2003;113(4):455–464.
13. Sarikcioglu L, Demir N, Demirtop A. A standardized method to create optic nerve crush: Yasargil aneurysm clip. *Exp Eye Res* 2007;84:373–377.
14. Tessema B, Pitman MJ, Roark RM, Berzofsky C, Sharma S, Schaefer SD. Evaluation of functional recovery of recurrent laryngeal nerve using transoral laryngeal bipolar electromyography (ToL EMG): a rat model. *Ann Otol Rhinol Laryngol* 2008;117:604–608.
15. Inagi K, Rodriquez AA, Ford CN, Heisey DM. Transoral electromyographic recordings in botulinum toxin-injected rat larynges. *Ann Otol Rhinol Laryngol* 1997;106:956–964.
16. Roark RM, Li JC, Schaefer SD, Adam A, De Luca CJ. Multiple motor unit recordings of laryngeal muscles: the technique of vector laryngeal electromyography. *Laryngoscope* 2002;112:2196–2203.

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